Data Engineer Task Assignment

The Data

You are given 2 csv files:

1. Groceries - member_id, date, item - contains information about each single item bought
2. Categories - item, category - contains a row per item that contains the item and the category hierarchy to which it belongs

You can download the files using these links:

1. https://deusdataengineering.blob.core.windows.net/data/groceries.csv
2. https://deusdataengineering.blob.core.windows.net/data/categories.csv

Groceries

Example rows:

```
3318,21-07-2015,other vegetables
3318,21-07-2015,curd
```

These rows mean that the member with id 3318 bought `other vegetables` and `curd` on the 21st of July 2015.

This counts as a cart - we suppose that all items bought in a single day by the same member were bought together.

Categories

Example row:

```
abrasive cleaner,Paper-Cleaning-Home/Cleaners-Supplies/All-Purpose-Cleaners
```

This row means that `abrasive cleaner` belongs to `All-Purpose-Cleaners`, which is a child of the `Cleaners-Supplies` category, which in turn is a child of the `Paper-Cleaning-Home` category.

Note: each item belongs in a single category, but the category itself might belong to another category. You can assume
that there can be up to 3 categories in each row.

Task Description

The assignment's goal is to load that data into a database. The database should contain:

1. All the items
2. All the categories
3. The item-category relationship
4. The member ids
5. The member_id - date-of-visit relationship
6. The item sales

The schema that you design should support the following queries:

1. Find the most frequent members
2. Find the most bought items
3. Find the biggest cart in the database (= the largest number of items bought together)
4. Find all the products under a category (e.g. "Meat-Seafood")
5. Find all the first-level children of a category (e.g. `Dairy-Eggs-Cheese`)

Technology Stack

You can implement this task in any language, framework and database of choice. We would appreciate it however if you used PostgreSQL and Pandas or PySpark. Feel free to over-engineer some parts to show-off your knowledge in anything that you use.

Deliverables

The deliverable must contain:
1. The ETL code that loads the data into the database
2. The queries mentioned above written in the query language supported by the database you chose
3. A readme file that documents what we need to do to run your code and produce the database